MIL-B-43655A

28 July 1975
SUPERSEDING
MIL-B-43655
9 July 1969 and
MIL-B-17470B
20 October 1967

MILITARY SPECIFICATION

BRUSHES; INJECTOR, CLEANING AND WIRE, TUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers round wire brushes (see 6.1).

* 1.2 Classification. The brushes, with the head diameters indicated, shall be of the following sizes, as specified (see 6.2).

Size 1 - 3/4 inch
Size 2 - 1 inch
Size 3 - 1/2 inch
Size 4 - 7/8 inch
Size 5 - 1/4 inch

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| H-S-951 | - | Synthetic Filaments for Brushes; General Specification for |
| QQ-W-321 | - | Wire, Copper Alloy |
| PPP-B-1729 | - | Brushes, Packaging and Packing of |

FSC 7920

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STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Standard product. The brushes covered by this specification shall be the manufacturer's current commercial product except for any changes necessary to comply with the requirements specified herein.

3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for brushes provided the completed item complies with all requirements specified herein.

* 3.2.1 Polyamide filler material. The polyamide filler material shall conform to class N, type III, .012 inch diameter of H-S-951.

* 3.2.2 Hog bristle. Hog bristle shall be pure black without artificial coloring and cleaned and sterilized as specified in 3.5 when tested as specified in 4.4.1. The hog bristle shall be treated so as to contain not less than 0.50 percent nor more than 0.70 percent DDT (dichloro-diphenyl-trichoroethane). The DDT content shall be based on the air dried weight of the treated bristle when tested as specified in 4.4.2. The hog bristle shall be sterilized after washing and rinsing, as follows. The temperatures and pressures shall be recorded by the supplier (see 4.2.1.3).

The hog bristle shall be soaked in boiling water (212°F.) for not less than 30 minutes.
The hog bristle shall be placed in a chamber and a temperature above 284°F. shall be maintained for not less than 3 hours.
The hog bristle shall be autoclaved at not less than 15 pounds per square inch steam pressure for not less than 45 minutes.

* 3.2.3 Copper alloy wire. The copper alloy wire shall be any composition, hard temper or better, round wire of QQ-W-321 having a diameter of 0.005 inch. Commercial tolerance shall apply.

* 3.3 Construction. The brushes shall have a zinc-coated steel, twisted wire, single spiral stem. One end of the stem shall extend beyond the brush head and form a handle terminating in a loop. The other end of the



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stem shall extend beyond the brush head, terminating in a blunt end. The head shall be formed by the even distribution of the filler material. The filler material filaments shall be parallel to each other and shall be at an angle of 90°F. to the stem. The brushes shall conform to table I. Filler material for the size 1 brush shall be the copper alloy wire specified in 3.2.3. Filler material for the size 2, size 3, size 4, and size 5 brush shall be either the polyamide material specified in 3.2.1 or the hog bristle specified in 3.2.2.

TABLE I. Minimum requirements

| | Size 1 | Size 2 | Size 3 | Size 4 | Size 5 |
|---|---|---|---|---|---|
| Length (inch) | | | | | |
| Overall | 21 | 12 | 12 | 9 | 6 |
| Brush head | 6 | 3-1/2 | 3-1/2 | 2-1/2 | 2-1/2 |
| Diameter of brush head (inch) | 3/4 | 1 | 1/2 | 7/8 | 1/4 |
| Total weight of filler material (ozs.) | 0.50 | .34 | .17 | .22 | .06 |
| Zinc coated steel wire diameter (inches commercial tolerance) | 0.1055 | 0.1055 | 0.0540 | 0.0625 | 0.0475 |
| Diameter of wire loop (inch) | 1 | 3/4 | 3/4 | 1/2 | 1/2 |

3.4 Workmanship. The filler material shall be clean and free of foreign material. The spiral twist in the stem shall retain the filler material in the brush head. The stem and handle shall be free from chips, blisters, voids or burrs, sharp edges in the zinc coating or basic wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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* 4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Components and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.2.1.1 Color of bristle. The bristle shall be tested in accordance with 4.4.1 for evidence of artificial coloring to determine compliance with 3.2.2. The sample shall be 2 ounces of bristle. The lot shall be all the bristles offered for inspection at one time. Any evidence of artificial coloring in the sample shall be cause for rejection of the lot.

* 4.2.1.2 DDT content. The hog bristle shall be tested in accordance with 4.4.2 to determine compliance with 3.2.2. The sample shall be 10 grams (g) $\pm$ 1 g of treated bristle. The lot shall be all bristle offered for inspection at one time. Failure of the sample to meet the DDT treatment requirement of 3.2.2 shall be cause for rejection of the lot.

* 4.2.1.3 Sterilization. Examination shall be made during the manufacturing process to determine that the sterilization of the hog bristle is in accordance with 3.2.2. Whenever nonconformance is noted correction shall be made to the lot affected and process.

* 4.2.2 End product inspection. A lot shall consist of all brushes of one size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated brush.

* 4.2.2.1 Visual examination. Examination shall be made of brushes for defects in table II. The inspection level shall be level I with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units.

TABLE II. Defects

| Examine | Defect |
|---|---|
| Brush head filaments | Imbedded foreign matter or not clean, metal wire stained |
| | Not held by stem |
| | Not parallel; not evenly distributed |
| Stem and handle | Not single stem single spiral |
| | Not provided with a loop |
| | Head end not blunt |
| | Any burr or sharp edge |
| | Void, chip, oil blister in zinc coating |
| | Handle bent |



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* 4.2.2.2 Dimensional examination. Inspection shall be made of the brush for defects in dimensions. Any dimensions not within specification requirements shall be classified as a defect. The inspection level shall be S-1 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

* 4.2.2.3 Testing of the end item. Sample brushes shall be tested in accordance with 4.4.3. The inspection level shall be S-2 with an AQL of 2.5 defects expressed in terms of defects per hundred units.

* 4.3.3 Examination of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing and marking comply with the requirements of PPP-B-1729.

4.4 Tests.

4.4.1 Color test. The sample of the hog bristle shall be boiled in 8 ounces of clear water at 212°F. for not less than 30 minutes, adding replacement water as required. Evidence of color in the water constitutes failure of this test.

4.4.2 DDT test. The sample unit shall be extracted for two hours in a Soxhlet apparatus with petroleum ether or any other suitable solvent, after which most of the solvent shall be removed by evaporation on a steam bath. Do not evaporate to dryness, since the DDT may decompose. Add 25 ml of 99 percent isopropyl alcohol and 2.5 g of metallic sodium cut into small pieces and swirl the flask in order to mix its contents. Connect to a water-cooled reflux condenser and boil gently for a minimum of 30 minutes. Shake the flask occasionally. Decompose the excess sodium by cautiously adding 10 ml of 50 percent isopropyl alcohol through the condenser at a rate of one to two drops per second. Boil for an additional 10 minutes and then add 60 ml of distilled water. Cool to room temperature, add two to three drops of phenolphthalein solution, neutralize by adding nitric acid (1 to 1), and add 10 ml of the diluted acid in excess. Add dropwise, with stirring of the solution a measured excess (25 ml) of 0.1N $AgNO_3$ solution. Coagulate the precipitate by heating on a steam bath for 30 minutes. Cool to room temperature and filter through a No. 52 Whatman filter paper and wash thoroughly with distilled water, receiving the filtrate in a 500-ml Erlenmeyer flask. Add 5 ml of ferric amonium sulfate indicator and titrate the excess $AgNO_3$ with 0.1N KSCN. Compute the net number of ml of 0.1N $AgNO_3$ consumed by the sample. Calculate percent DDT as follows:

$$\text{Percent DDT (dichloro-diphenyl-tricholorethane)} = \frac{\text{Milliliters of 0.1N } AgNO_3 \text{ (consumed) x .7094}}{\text{Weight of sample (grams)w}}$$

A blank determination for the chemicals used shall be made following the exact procedure given above, but limiting the 0.1N $AgNO_3$ solution to 5 ml to obtain a chloride correction value. This blank value shall be subtracted from the DDT determination to obtain the exact value of DDT present. Nonconformance to the requirement in 3.2.2 constitutes failure of this test.



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4.4.3 Weight of filler material.

4.4.3.1 Weight of polyamide or copper alloy filler material. The polyamide or copper alloy filler material shall be removed from the brush and weighed. Nonconformance to the weight requirement of table I constitutes failure of this test.

4.4.3.2 Weight of hog bristle. The bristle shall be removed from the brush and conditioned for not less than 16 hours in air at a relative humidity of 50 ± 4 percent and at a temperature of 23°C. The bristle shall be weighed. Nonconformance to the weight requirement in table I constitutes failure of this test.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for level A, B, or C shall be in accordance with PPP-B-1729.

6. NOTES

6.1 Intended use. The brushes covered by this specification are intended for use in cleaning injector tubes. The size 1 brush is intended to be used in cleaning dishwasher spray tubes. Size 2 and size 3 brushes are intended for use on industrial machines. Size 4 and size 5 brushes are intended for use on diesel engines during overhaul.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1).

* 6.3 Classification comparison. MIL-B-17470B has been included in this revision as follows:

| MIL-B-43655A | MIL-B-43655 | |
|---|---|---|
| Size 1 | - | MIL-B-17470B |
| Size 2 | Type I, Size 1 | - |
| Size 3 | Type I, Size 2 | - |
| Size 4 | Type II, Size 1 | - |
| Size 5 | Type II, Size 2 | - |

* 6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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